Alex Seal's Personal
Level 1 High Power Rocketry Project
"Figure.09"



Figure 1 – Low altitude burn image



Figure 2 – Alex Seal himself setting up his ignitor

Certification Organization: Tripoli
Project Duration: May 2026 – September 2026
Motor Used: CTI H225-14A White Thunder Single-Use
Certification Date: September 26th, 2026; Granted a Tripoli Level 1 Certification

Skills Demonstrated

Aerospace Vehicle Design • CAD Modeling (Autodesk Inventor) • Technical Drawing Creation • Flight Simulation (OpenRocket) • Stability Analysis • Manufacturing & Assembly • Systems Integration • Technical Documentation • Engineering Decision Making • Flight Testing & Verification

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- **Executive Summary**

This project involved the design, simulation, assembly, testing, and successful flight of a Level 1 high-power rocket. The primary objective was to obtain a Level 1 certification while applying engineering principles related to flight stability, structural integration, recovery system design, and launch operations.

The project included vehicle modeling using OpenRocket, flight performance analysis, CAD design and technical drawings, construction of the airframe and recovery systems, pre-flight safety inspections, and launch execution. The rocket successfully completed its intended flight profile and was recovered without damage, resulting in successful Level 1 certification.

- **Project Background**

Motivation

As a Mechanical Engineering student with aspirations to enter the aerospace industry, I sought opportunities to gain aerospace-specific experience beyond the classroom. While TCU does not offer a dedicated Aerospace Engineering program, I viewed that as an opportunity to demonstrate initiative rather than a limitation. By leveraging experiences such as SAE Aero and this Level 1 high-power rocket project, I sought to develop practical skills in flight systems development. Through independently planning, funding, and executing the project, I applied engineering concepts from my freshman year while gaining hands-on experience with simulation, subsystem integration, manufacturing, flight operations, technical documentation, and project management. The result was a complete aerospace system taken from concept to successful flight and recovery.

Project Objectives

- Obtain Level 1 High-Power Rocket Certification
- Design and assemble a flight-ready vehicle
- Validate flight performance through simulation
- Demonstrate stable flight characteristics
- Successfully deploy and recover the vehicle
- Document the complete engineering process

- **Research and Planning**

Since this project was starting from the ground up, research was extensive not only behind how to begin manufacturing a rocket, but also the logistics behind safely (and legally!) launching a high-power rocket.

Key planning activities included:

- Reviewing certification requirements and constraints, along with launch day procedures such as registration and location, which led to selecting a Tulsa Rocketry launch event as the certification venue on 9/26/2026
- Part and assembly research, including subsystems such as propulsion and recovery
- Learning OpenRocket simulation methods
- Developing a project schedule
- Observing successful Level 1 attempts and their designs, just as much as observing unsuccessful Level 1 attempts
- Part and assembly research

This phase resulted in more specified knowledge behind high-power rocketry, along with giving me a deadline that I could now revolve a schedule around. Which allowed me to move directly into my design process.

In addition, I had selected all my components that I could pull geometry from to complete a preliminary design.

- **Vehicle Design and Simulation**

Vehicle Configuration

Following the research phase, a preliminary vehicle configuration was developed to satisfy all Level 1 criteria while providing a platform that could be manufactured and customized for my own objectives. Rather than just following instructions on a kid, the rocket was designed with methodically selected components, and custom fin geometry and configuration.

Vehicle Specifications

NOTE - Not final values, rough values that all needed to be accounted for before purchasing of materials, apogee and stability values could only be so accurate without precise values for simulation

Parameter	**Value**
Height	47.8”
Diameter	4”
Launch Mass	~4.3 lb (estimated preliminary mass)
Motor	H-100 originally, but switched to H-225 (ran simulations with an H-225)
Predicted Apogee	1478 ft (451 m)
Stability Margin	1.68 Calibers

- *Values through OpenRocket check figure 4*

Preliminary Design

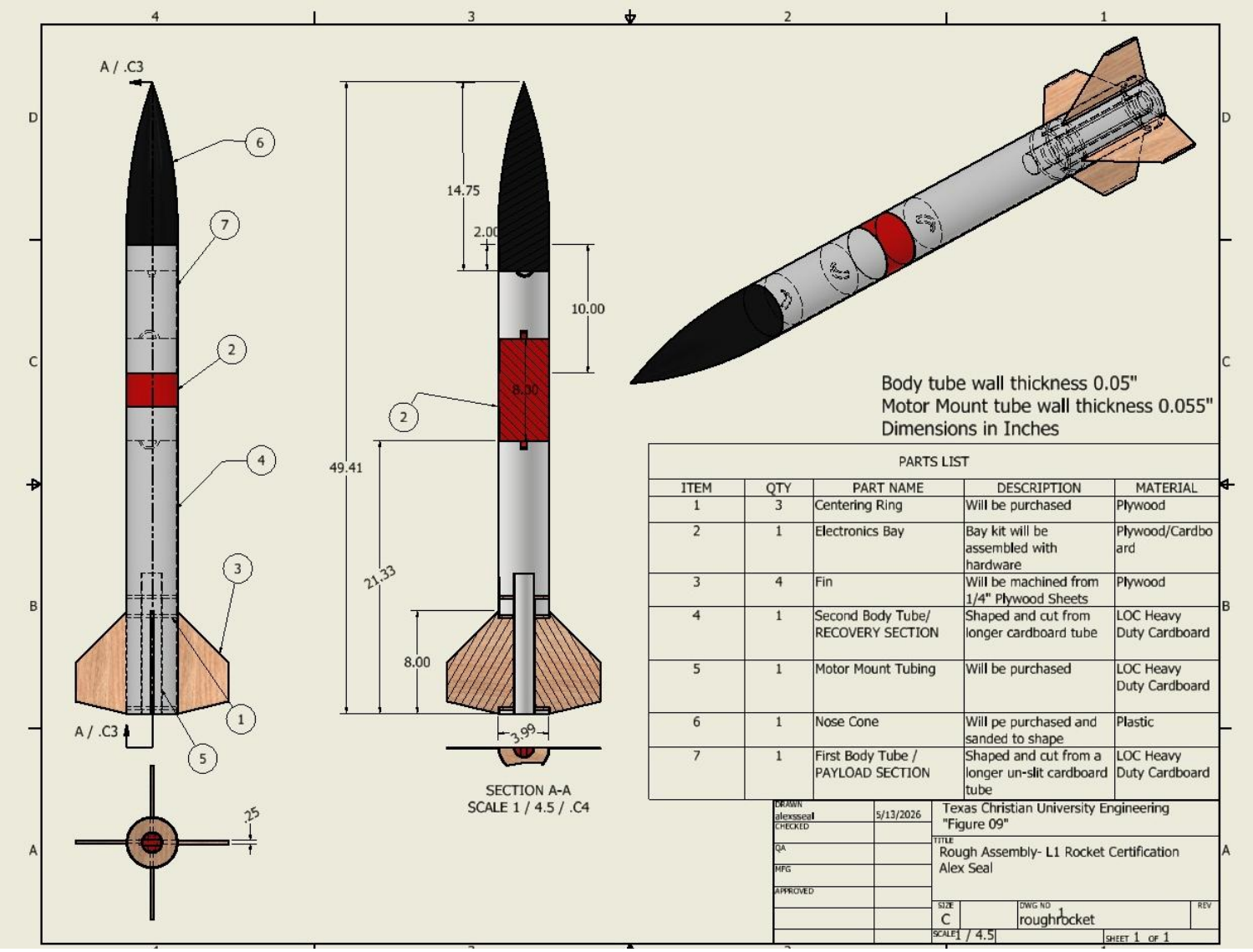


Figure 3 – CAD technical drawing of preliminary design

Flight Simulation

OpenRocket simulations were used to model vehicle performance and verify stable flight characteristics prior to launch.

Simulation objectives included:

- Stability verification
- Apogee prediction (Crucial for Recovery deployment)
- Velocity estimation

- Recovery performance assessment
- Motor suitability analysis

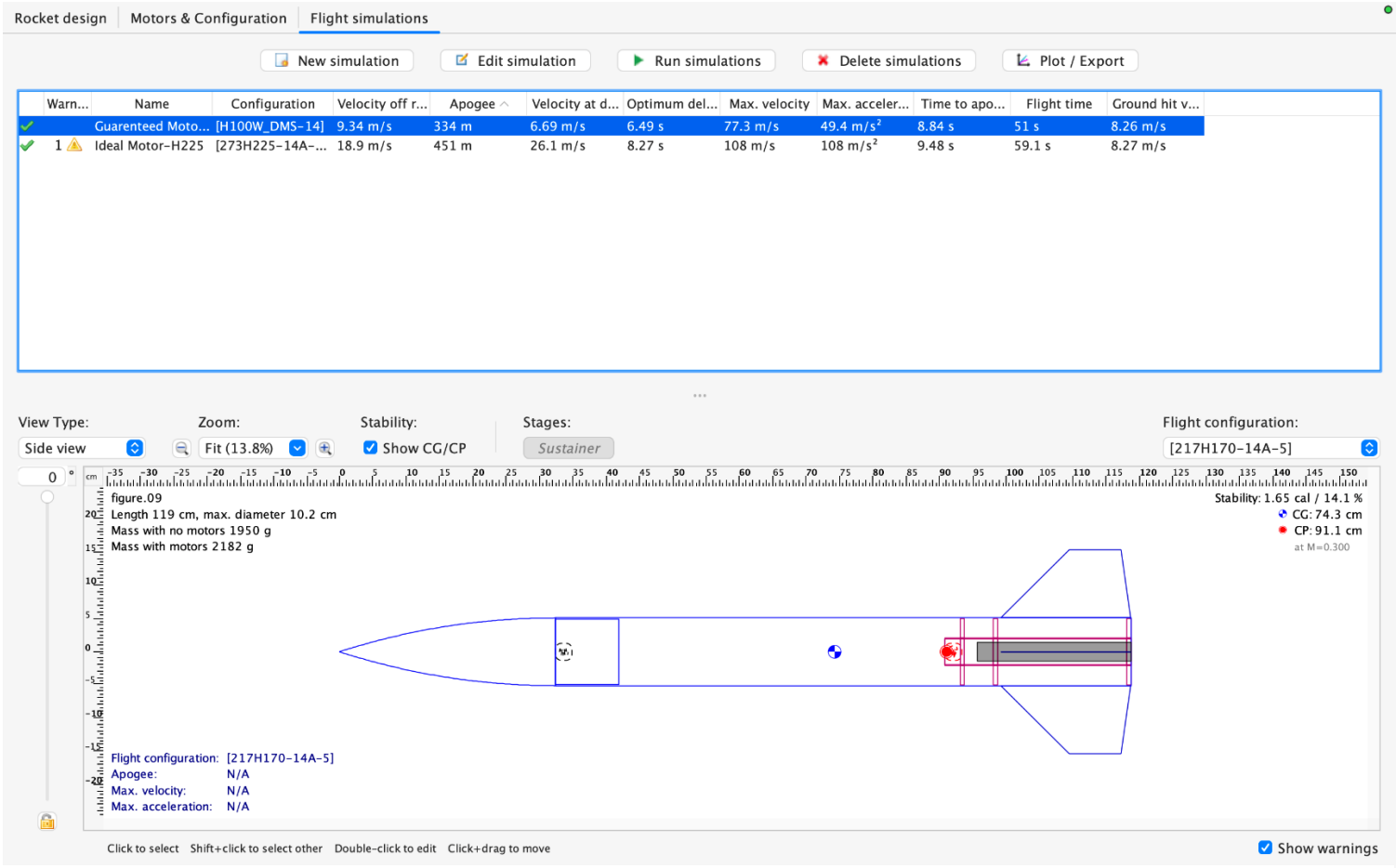


Figure 4 – OpenRocket Simulation

➔ Engineering Decisions ; MOTOR and RECOVERY

Several design decisions were made throughout the project to improve reliability, safety, and flight performance. For example, the size/strength of motor that I would be using, along with configuring my recovery system and parachute constraints.

I settled on a 36” LOC precision parachute and an 18ft 5/8” nylon shock cord that would hold my 2 halves of my rocket together, along with the swivel that is tied to my parachute. This gave me plenty big of a chute to create enough drag to slow my descent velocity to a safe level, and enough rope for the 2 halves to not crash into each other mid-flight. This would all be deployed by the adjustable charge that is built into the rocket motor that pops the top half of the rocket off with a small explosion. This charge is 14 seconds but can (and should) be adjusted to the height of your apogee. Recovery must be delivered at/close to apogee since that’s where the rocket is closest to rest as deployment at high speeds can shred your recovery equipment.

- Due to difficulty in obtaining rocket motors as an unlicensed flyer, I settled on an h100 that I secured through a rocket salesman that would be attention the Tulsa launch. However, after plugging everything into my OpenRocket simulation, it showed many issues.
    - Barely had an estimated thrust to weight ratio of 5:1 lbs, placing the vehicle on the lower limit of recommended launch safety margins
    - Didn't have enough exit velocity off a standard 6ft rail but only off a 12ft rail

Yes, this aircraft would technically be flyable, but operating so close to the minimum safety margins would present a wider possibility of flight errors and increased chance of weathercocking, in addition to just having a lower factor of safety.

- After comparing additional motor options, I settled on a H225 motor that gave me plenty of exit velocity and an almost 10:1 Thrust to weight ratio that increased the factor of safety, but also my factor of stability.

Decision	**Alternatives Considered**	**Final Choice**	**Reason**
Vehicle Construction	Commercial Kit	Custom built Rocket	Increased engineering involvement and allowed full control over vehicle design.
Airframe Diameter	3-5" airframe	4" airframe	Balanced aerodynamic performance, stability, and internal component space
Fin Configuration	3 fins, 4 fins	4 fins	Improved stability margin and provided a robust fin design
Fin Attachment Method	Surface mount fins, TTW fins	Through the wall fins	Increased structural integrity and load transfer into the airframe
Propulsion System	H100, H225	Cesaroni Technology Incorporated H225-14A motor	Provided higher launch rail exit velocity and improved flight performance
Recovery System	Smaller Parachute sizes	36" parachute	Produced a safe descent rate while minimizing landing loads

Manufacturing and Assembly

Following completion of the design and simulation phases, manufacturing and assembly of the vehicle began. Rather than utilizing a commercial kit, the rocket was constructed from individually selected components, including the airframe, nose cone, motor mount tubing, and electronics bay. Custom fin geometry developed during the design phase was transferred to plywood stock, cut to shape, and manually airfoiled to improve aerodynamic performance. Through-the-wall fin slots were then created in the airframe, allowing the fin structure to bond directly to the motor mount assembly and provide increased structural integrity. Following fin installation, the electronics bay, recovery system, and remaining airframe components were assembled and integrated to create a flight-ready vehicle. Structural fillets were formed with BSI epoxy thicken with fumed silica to reach desired viscosity for fillet shaping, increasing joint strength.

Electronics Bay Assembly

Purchased the LOC precision Electronics bay and practiced by epoxy application skills. Electronics bay not only served well for any electronics on flight but allowed me to divide my rocket into 2 sections, the recovery section, and payload section.

Fin Geometry Manufacturing

Fin Geometry was a simple yet very crucial step in this rocket's design. I settled on an ARCAS fin shape, which is a square and a triangle with side lengths equal to the rocket's diameter the bottom edges were tapered up 15 degrees as the sharp corner makes contact with the ground very quick that risks damage. I also settled on a 4 fin configuration. Fins were cut out of pre machined ¼" plywood panels with an electric jigsaw by hand. Fins were also sanded given an airfoiled shape to reduce drag, but not sanded to less than 1/8" tip to prevent splintering. On top of the shape a "tab" was cut into the shape as well with a length of , which gave 3 points of contact with the centering rings that allowed for a traditional "through the wall" fin design.

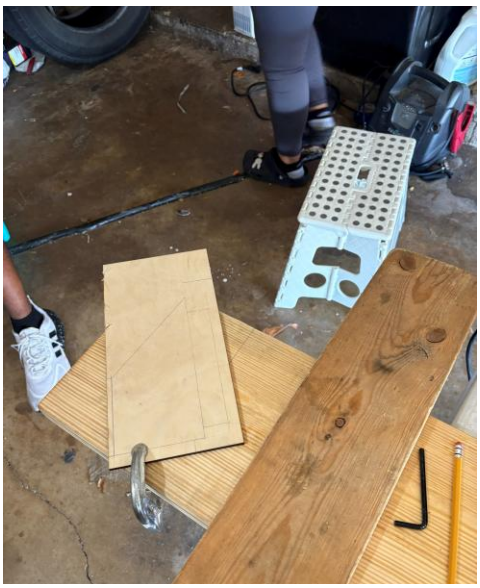


Figure 5- Geometry drawn on plywood stock



Figure 6 – Airfoil integration to reduce drag

Fin Can Assembly

LOC precision centering rings were epoxied onto motor mount tubing, with the bottom (aft) centering ring and the middle ring being spaced exactly the height of the "tabs" that were cut into the fins. They were epoxied into place and spaces exactly 90 degrees apart from each other with printed templates and 3D printed clocking sleeve (ensuring fins weren't tilted) and then epoxy fillets were added in all corners to reinforce the structure and close all gaps. An eye hook was added into the top centering ring to give recovery shockcord an eyelet to hook onto. T nuts were also placed in the aft ring for motor retention.



Figure 7- Epoxy fillets from motor mount tube to fins

Sealing the Rocket Off

Once the fin can was finalized, quarter inch slits were cut 90 degrees apart from each other, 8 inches into the body tube. Dry fit was tested and once the fit was clean, a second round of epoxy fillets were added for a smooth and strong fin to body transition, sealing off the bottom end of the rocket. Once the rocket was sealed, a 10" section was divided from the body tube to make the aforementioned payload section. This section is friction tightened to the E-bay while the bottom half is a bit looser and it has to come apart during the recovery event. I purchased a jollylogic altimeter which went into my payload section to record my final height.

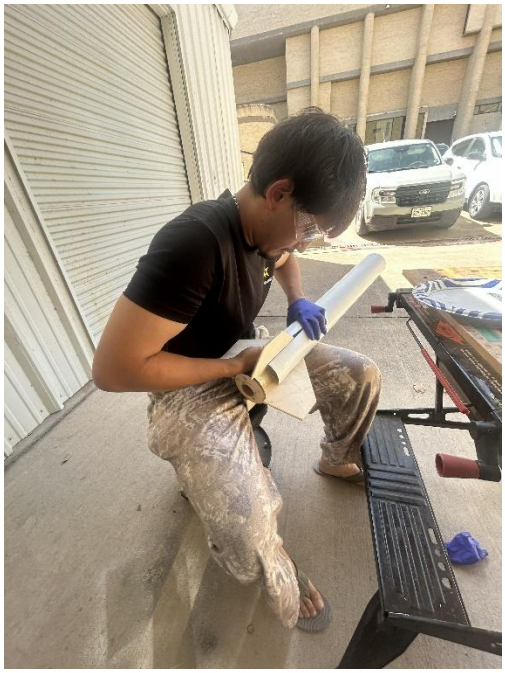


Figure 8 – Alex sliding epoxy covered fin can into place



Figure 9 – Exterior epoxy fillets curing



Figure 10 – Final shape of the aft end of rocket

Vehicle Finishing and Appearance



Figure 11 – Spray painted

As the project progressed from concept to flight-ready vehicle, I wanted the final product to reflect the level of effort invested in its design and construction. The rocket was finished in a black and gold color scheme inspired by the song *Figure.09*. Beyond aesthetics, this personalization helped establish a unique project identity and reinforced the satisfaction of creating a vehicle from individually selected, designed, and manufactured components.

Recovery System Integration

As previously mentioned, the bottom half of the rocket hosted the recovery system. These rocket motors produce a hot of heat so after folding my parachute, I wrapped it in an orange kevlar heat blanket and also wrapped a blanket around the entire shock cord. In addition to all this, I packed around 5 inches of rocketry recovery wadding that takes the blunt of the secondary recovery charge. The two halves of the rocket were tied with the 18' cord, and I tied a knot 12 ft away from the top half of the rocket, where I connected my parachute and swivel. This specific configuration allows the two halves to not collide middair due to the different lengths. All quick links were thread locked to ensure nothing would come undone during launch.



Figure 12 – Recovery system configuration

- Shock cord
- Recovery heat protection (blankets and wadding)
- Parachute
- Attachment hardware

Final Assembly

Following subsystem integration, the vehicle underwent final assembly and inspection procedures.

Inspection items included:

- Structural integrity
- Recovery system functionality
- Motor retention
- Launch lug/rail guide verification
- Stability confirmation

From here I could also record my final mass, to do my final updated flight simulation for the most accurate values possible. My rockets final mass came out to 4.625 lbm without a motor.



Figure 13 – Final Assembly



Figure 14 – Ignitor set up

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Launch Day

Pre-Launch Activities

Prior to launch, the vehicle was inspected to verify flight readiness and compliance with certification requirements.

Activities included:

- Motor installation
- Recovery system packing
- Safety inspection
- Weather assessment
- Launch pad setup
- Adjusting Recovery Charge Delay

After simulating my rockets flight with the true mass, I calculated my apogee to be at 9.23 seconds. So I adjusted by 14 second delay by 4 seconds, giving me a deployment delay time of 10 seconds after launch. This means my recovery event is 10 seconds after launch when the rocket should be close to stationary.

Launch

The rocket was launched using a H225 motor under suitable weather conditions.

Observations during powered ascent:

- Stable liftoff
- Straight flight path
- Nominal motor burn
- Successful coast phase



Figure 1

Recovery

Recovery deployment occurred as expected and the vehicle returned safely to the ground.

Recovery observations:

- Successful parachute deployment
- Controlled descent
- No significant damage
- Vehicle suitable for future flights

Results and Performance Assessment

Flight Outcome

The flight successfully met all primary mission objectives, had a stable flight, recovery event occurred and rocket split in 2 whilst staying together to be guided down by the parachute that was successfully installed.

Checklist

- Flyable condition
- Recovery system didn't burn
- No damage
- Approved by Tulsa Rocketry Prefact



Figure 15 – Altimeter reading after recovering rocket

Predicted vs Actual Performance

Parameter	Predicted	Actual

Altitude	1377 ft	1310 ft
Apogee	9.32 s	9.0s
Flight Time	53.3 s	48.5 s
Recovery Distance	N/A	300 ft

➔ The final flight served as a design verification test, which allowed me to compare simulated and measured performance metrics.

The measured altitude resulted in a percent error of 4.87%, which is a significant amount caused by high fin drag, and from the fact that the rocket took off at a slight angle meaning part of it's trajectory is horizontal instead of purely vertical.

Our flight time returns a percent error of 9.01% which can be factoring from the recovery system and wind conditions.

Our apogee returns a percent error of 43% which may have also been influenced by wind conditions and slight variations in the behavior of the motors single stage ignition.

Lessons Learned

This project provided valuable experience in aerospace systems engineering, launch operations, and technical decision-making.

Key lessons learned include:

- Importance of recovery system preparation
- Benefits of simulation-based planning, simulation and verification give you another set of eyes.
- Effects of mass distribution on vehicle stability
- Importance of pre-flight inspections
- Value of thorough documentation
- Launch operations and range safety procedures

Future Improvements

Potential future enhancements include:

- Flight data logging
- Composite airframe construction
- Higher-altitude flight profiles

- Better documentation of the manufacturing process

Conclusion

This project successfully met the objective of designing, manufacturing, and launching a custom high-power rocket to achieve Tripoli Level 1 certification.

From start to finish, I was able to incorporate engineering principles learned in school, along with manufacturing principals learned from design projects into one concise project. Unlike a typical kit-based build, this rocket was designed from individually selected components and incorporated custom airfoiled fins with assembly configuration that prioritized structural integrity.

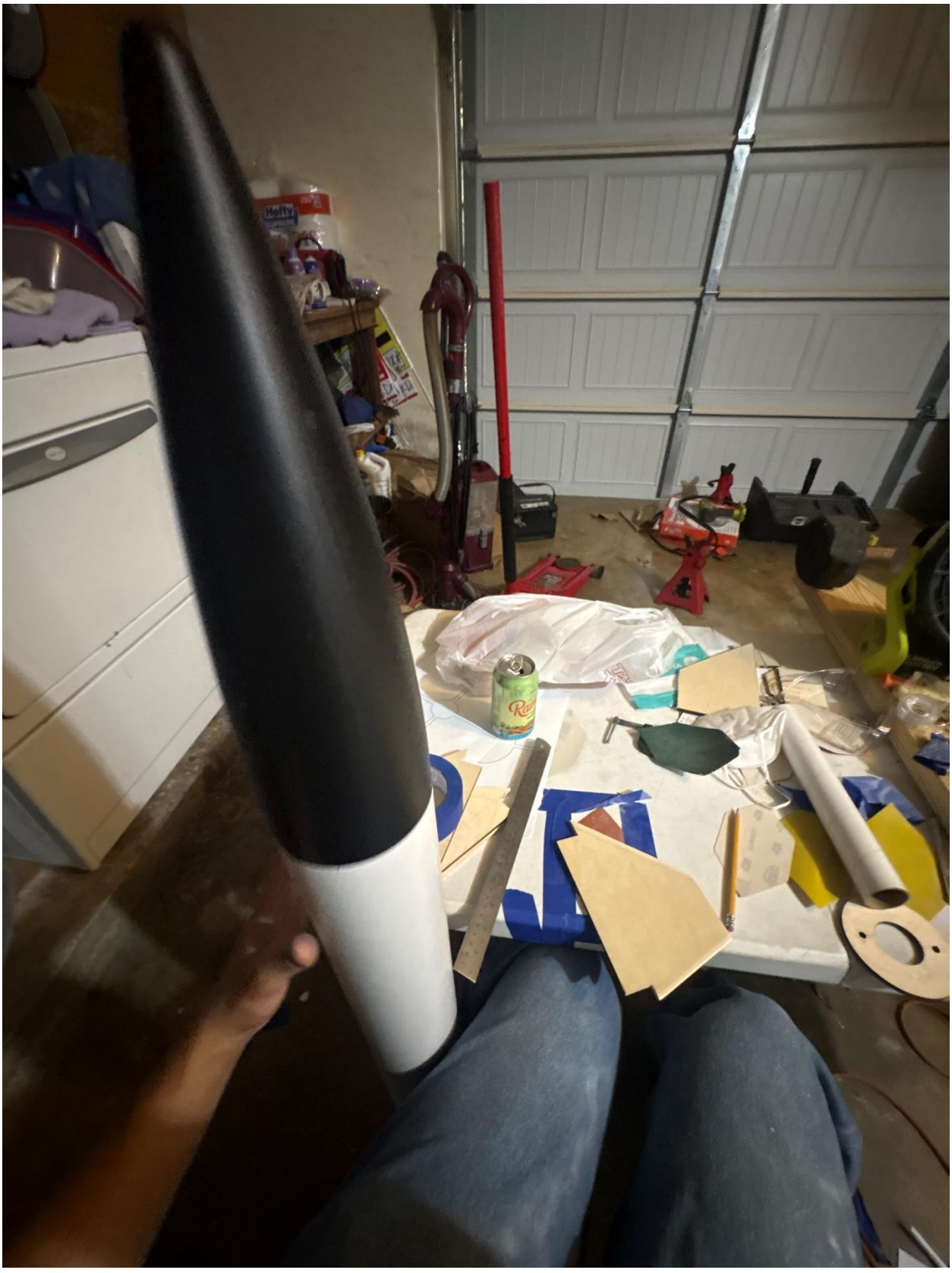
Simulation performed a huge role in the engineering process, as simulations guided several engineering decisions, most notably the selection of the CTI H225 motor, which provided enough thrust for a stable rail exit velocity. This notes the advantage of a simulation based design. In addition, I was able to build on my technical CAD and technical drawing creation skills, flight simulation, systems integration, demonstrating the complete development cycle of an aerospace vehicle. The project required requirements evaluation, simulation-driven design, manufacturing, systems integration, testing, verification, and flight operations, providing practical experience with the complete engineering development process.

The completed design was able to reach a final height of 1310 ft in the sky while returning back without even a scratch. The rocket “Figure 09” met all requirements, allowing me to receive my Level 1 Rocket Certification. This certificate now allows me to buy rocket motors which I hope to use to start a rocket propulsion club at Texas Christian University to expand the presence of design teams.

Appendix A - Additional Images

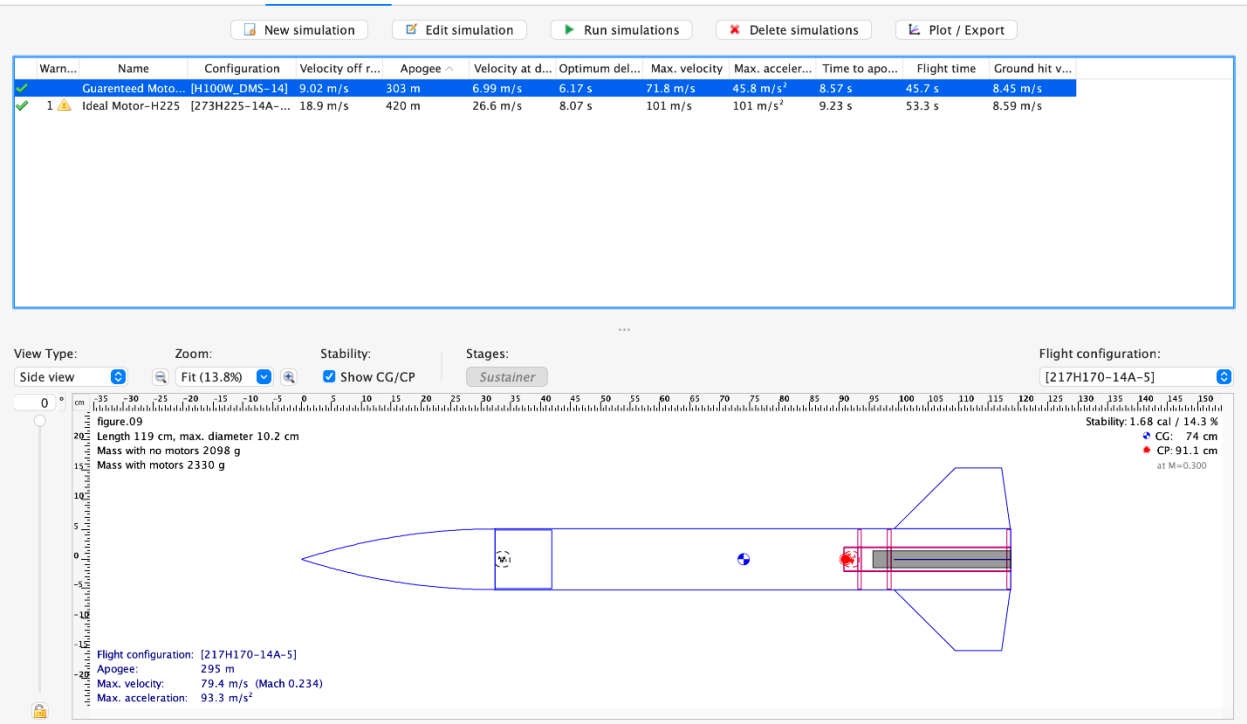

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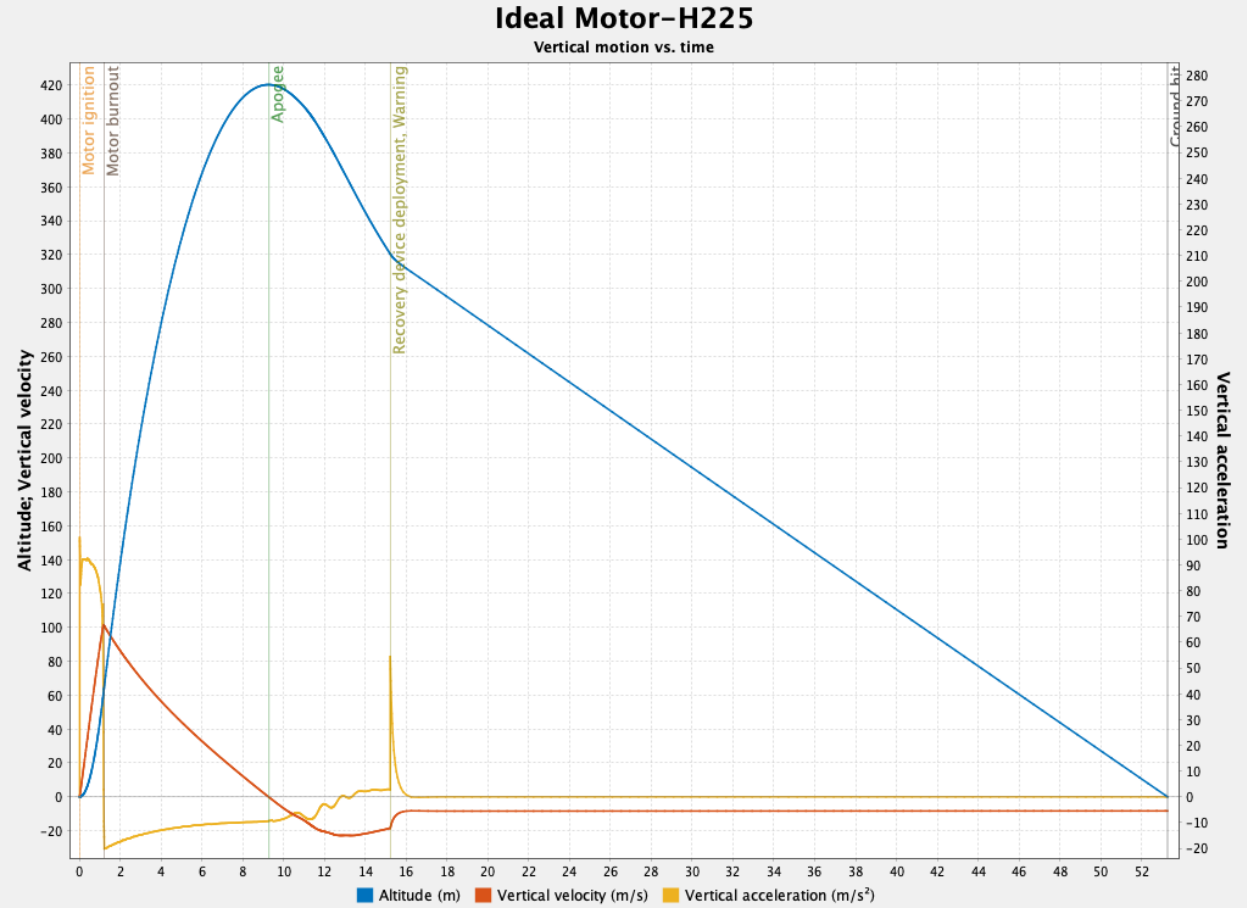


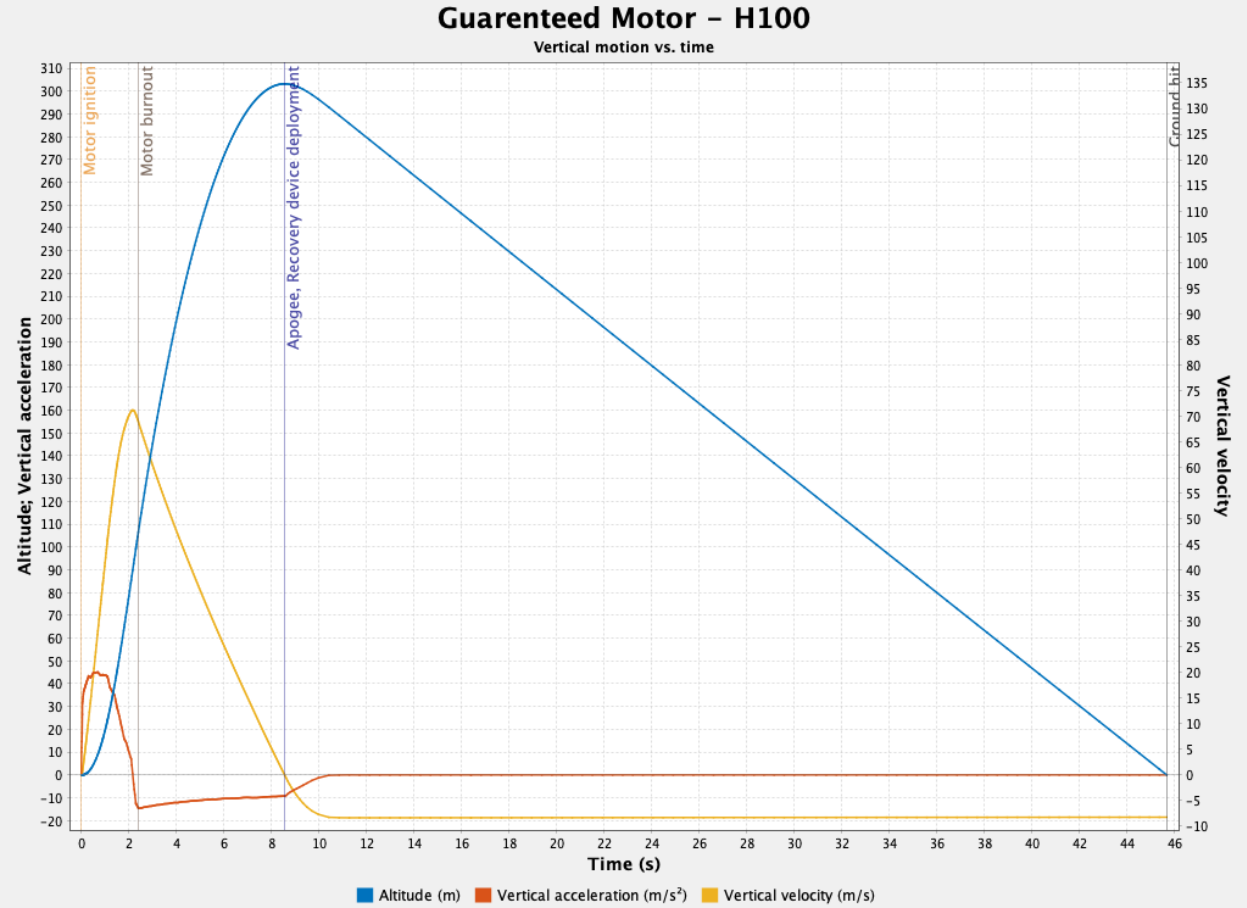

Hefty

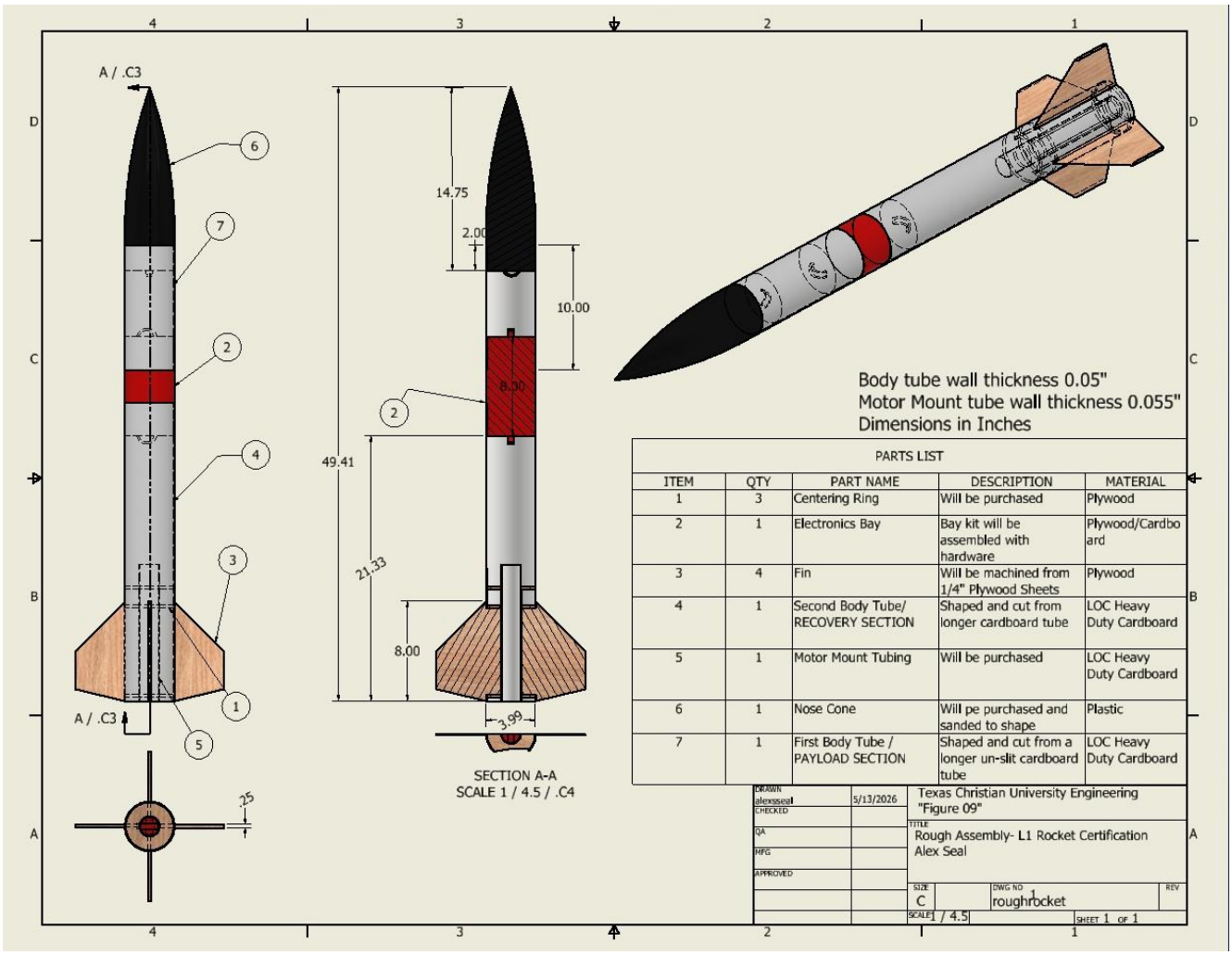


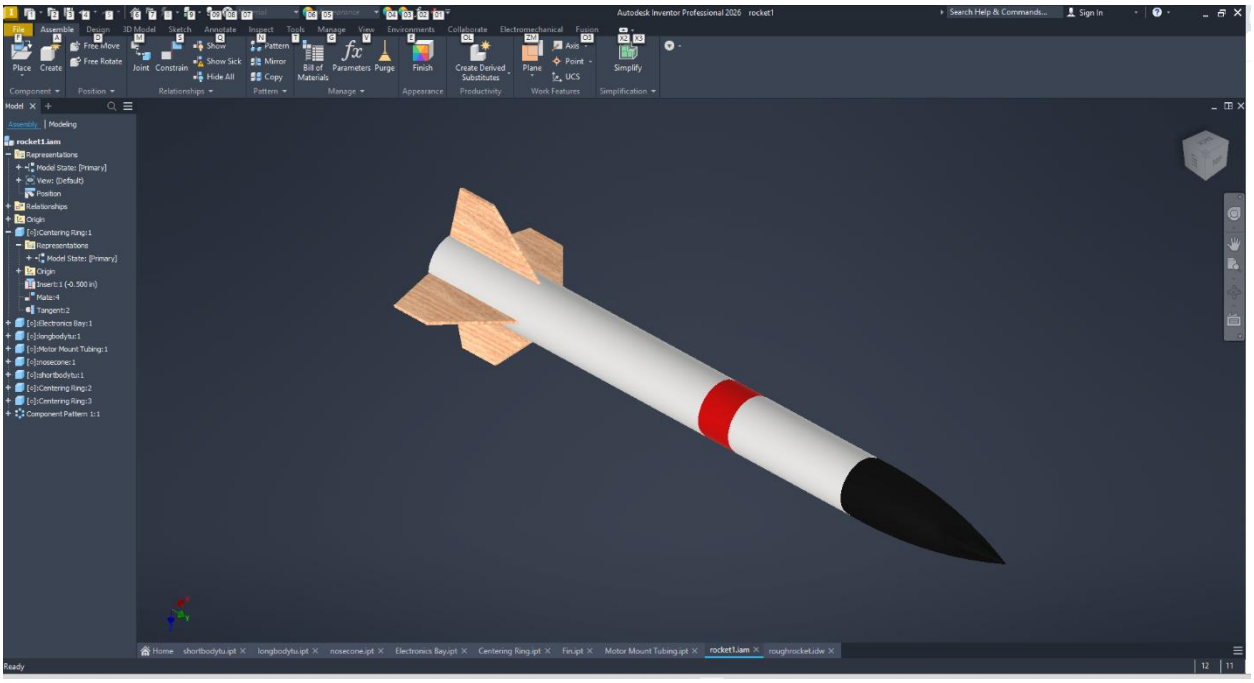
Appendix B - Simulation Data for final rocket simulations with a H225 and an H100 AND CAD


New simulation
Edit simulation
Run simulations
Delete simulations
Plot / Export
Warn... Name Configuration Velocity off r... Apogee Velocity at d... Optimum del... Max. velocity Max. acceler... Time to apo... Flight time Ground hit v...
Guarenteed Moto... [H100W_DMS-14] 9.02 m/s 303 m 6.99 m/s 6.17 s 71.8 m/s 45.8 m/s² 8.57 s 45.7 s 8.45 m/s
1 Ideal Motor-H225 [273H225-14A-... 18.9 m/s 420 m 26.6 m/s 8.07 s 101 m/s 101 m/s² 9.23 s 53.3 s 8.59 m/s
View Type:
Side view
Zoom:
Fit (13.8%)
Stability:
Show CG/CP
Stages:
Sustainer
Flight configuration:
[217H170-14A-5]
figure.09
Length 119 cm, max. diameter 10.2 cm
Mass with no motors 2098 g
Mass with motors 2330 g
Stability: 1.68 cal / 14.3 %
CG: 74 cm
CP: 91.1 cm
at M=0.300
Flight configuration: [217H170-14A-5]
Apogee: 295 m
Max. velocity: 79.4 m/s (Mach 0.234)
Max. acceleration: 93.3 m/s²


Ideal Motor-H225
Vertical motion vs. time
Motor ignition
Motor burnout
Apogee
Recovery device deployment, Warning
Ground hit
Altitude; Vertical velocity
Vertical acceleration
Altitude (m)
Vertical velocity (m/s)
Vertical acceleration (m/s²)


Guarenteed Motor – H100
Vertical motion vs. time
Motor ignition
Motor burnout
Apogee, Recovery device deployment
Ground hit
Altitude; Vertical acceleration
Vertical velocity
Time (s)
Altitude (m)
Vertical acceleration (m/s²)
Vertical velocity (m/s)


A / .C3
6
7
2
4
3
1
5
A / .C3
14.75
2.00
10.00
8.30
2
49.41
21.33
8.00
3.99
.25
SECTION A-A
SCALE 1 / 4.5 / .C4
Body tube wall thickness 0.05"
Motor Mount tube wall thickness 0.055"
Dimensions in Inches
PARTS LIST
ITEM | QTY | PART NAME | DESCRIPTION | MATERIAL
1 | 3 | Centering Ring | Will be purchased | Plywood
2 | 1 | Electronics Bay | Bay kit will be assembled with hardware | Plywood/Cardboard
3 | 4 | Fin | Will be machined from 1/4" Plywood Sheets | Plywood
4 | 1 | Second Body Tube/ RECOVERY SECTION | Shaped and cut from longer cardboard tube | LOC Heavy Duty Cardboard
5 | 1 | Motor Mount Tubing | Will be purchased | LOC Heavy Duty Cardboard
6 | 1 | Nose Cone | Will pe purchased and sanded to shape | Plastic
7 | 1 | First Body Tube / PAYLOAD SECTION | Shaped and cut from a longer un-slit cardboard tube | LOC Heavy Duty Cardboard
DRAWN alexisseal 5/13/2026
CHECKED
QA
MFG
APPROVED
Texas Christian University Engineering "Figure 09"
TITLE Rough Assembly- L1 Rocket Certification Alex Seal
SIZE C
DWG NO roughrocket
REV
SCALE 1 / 4.5
SHEET 1 OF 1

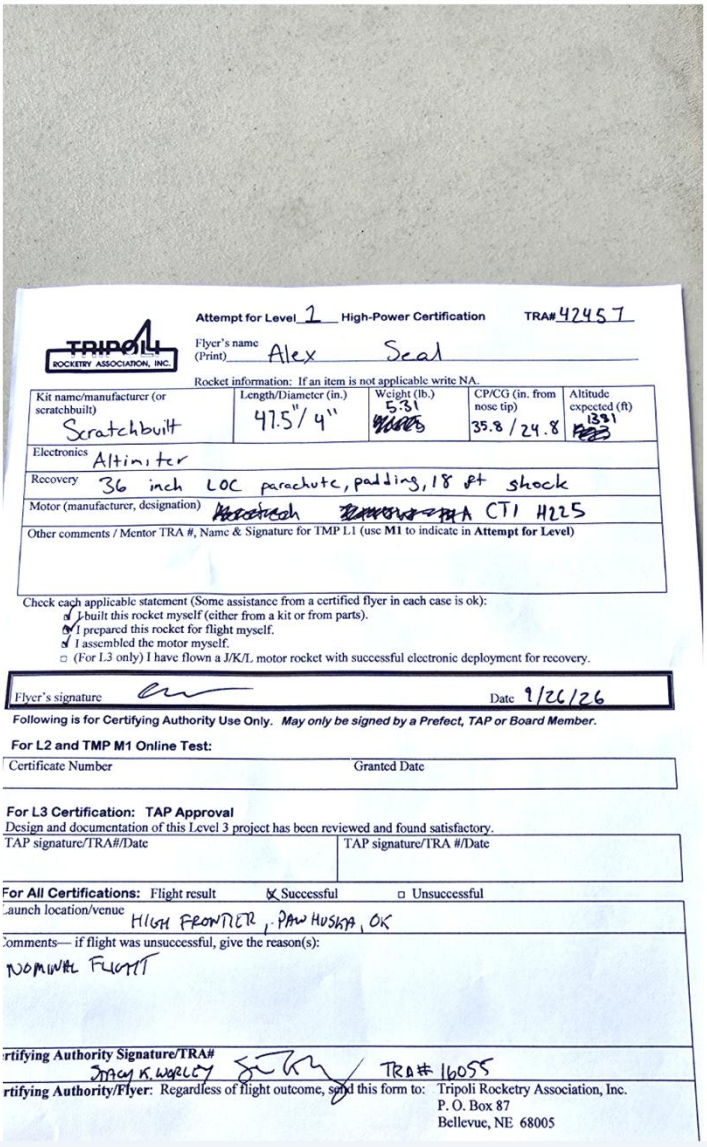


Appendix C - Certification Documentation



Please process the attached successful L1 certification flight.

Thanks!



TRIPOLI ROCKETRY ASSOCIATION, INC.

Attempt for Level 1 **High-Power Certification** **TRA#** 42457

Flyer's name (Print) Alex Seal

Rocket information: If an item is not applicable write NA.

Kit name/manufacturer (or scratchbuilt)	Length/Diameter (in.)	Weight (lb.)	CP/CG (in. from nose tip)	Altitude expected (ft)
Scratchbuilt	47.5" / 4"	5.31	35.8 / 24.8	1331

Electronics Altimiter

Recovery 36 inch LOC parachute, padding, 18 ft shock

Motor (manufacturer, designation) [illegible] CTI H225

Other comments / Mentor TRA #, Name & Signature for TMP L1 (use **M1** to indicate in **Attempt for Level**)

Check each applicable statement (Some assistance from a certified flyer in each case is ok):
- ☑ I built this rocket myself (either from a kit or from parts).
- ☑ I prepared this rocket for flight myself.
- ☑ I assembled the motor myself.
- ☐ (For L3 only) I have flown a J/K/L motor rocket with successful electronic deployment for recovery.

Flyer's signature [signature] Date 9/26/26

Following is for Certifying Authority Use Only. ***May only be signed by a Prefect, TAP or Board Member.***

For L2 and TMP M1 Online Test:

Certificate Number	Granted Date

For L3 Certification: TAP Approval

Design and documentation of this Level 3 project has been reviewed and found satisfactory.

TAP signature/TRA#/Date	TAP signature/TRA #/Date

For All Certifications: Flight result ☒ Successful ☐ Unsuccessful

Launch location/venue HIGH FRONTIER, PAW HUSKA, OK

Comments— if flight was unsuccessful, give the reason(s):

NOMINAL FLIGHT

rtifying Authority Signature/TRA# Stacy K. Worley [signature] TRA# 16055

rtifying Authority/Flyer: Regardless of flight outcome, send this form to: Tripoli Rocketry Association, Inc.
P. O. Box 87
Bellevue, NE 68005